

Activity Demands During Multi-Directional Team Sports: A Systematic Review

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Published online: 12 August 2017
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Abstract

Background Late-stage rehabilitation programs often incorporate ‘sport-specific’ demands, but may not optimally simulate the in-game volume or intensity of such activities as sprinting, cutting, jumping, and lateral movement.

Objective The aim of this review was to characterize, quantify, and compare straight-line running and multi-directional demands during sport competition.

Data Sources A systematic review of PubMed, CINAHL, SPORTDiscus, and Cochrane Central Register of Controlled Trials databases was conducted.

Study Eligibility Criteria Studies that reported time-motion analysis data on straight-line running, accelerations/decelerations, activity changes, jumping, cutting, or lateral movement over the course of an entire competition in a multi-directional sport (soccer, basketball, lacrosse, handball, field hockey, futsal, volleyball) were included.

Study Appraisal and Synthesis Methods Data was organized based on sport, age level, and sex and descriptive statistics of the frequency, intensity, time, and volume of the characteristics of running and multi-directional demands were extracted from each study.

Results Eighty-one studies were included in the review ($n = 47$ soccer, $n = 11$ basketball, $n = 9$ handball, $n = 7$ field hockey, $n = 3$ futsal, $n = 4$ volleyball). Variability of sport demand data was found across sports, sexes, and age levels. Specifically, soccer and field hockey demanded the most volume of running, while basketball required the highest ratio of high-intensity running to sprinting. Athletes change activity between 500 and 3000 times over the course of a competition, or once every 2–4 s. Studies of soccer reported the most frequent cutting (up to 800 per game), while studies of basketball reported the highest frequency of lateral movement (up to 450 per game). Basketball (42–56 per game), handball (up to 90 per game), and volleyball (up to 35 per game) were found to require the most jumping.

Limitations These data may provide an incomplete view of an athlete’s straight-line running load, considering that only competition and not practice data was provided.

Conclusions Considerable variability exists in the demands of straight-line running and multi-directional demands across sports, competition levels, and sexes, indicating the need for sports medicine clinicians to design future rehabilitation programs with improved specificity (including the type of activity and dosage) to these demands.

Electronic supplementary material The online version of this article (doi:[10.1007/s40279-017-0772-5](https://doi.org/10.1007/s40279-017-0772-5)) contains supplementary material, which is available to authorized users.

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Key Points

This review provides reference data for sports medicine clinicians and strength and conditioning professionals to use during the design of performance or rehabilitation programs to determine running progressions, jump counts or interval cutting programs that most systematically progress athletes and help with return-to-play decisions.

There were substantial differences in straight-line running, cutting, jumping, and lateral movement between sports, ages, and sexes.

Soccer and field hockey demand the largest total running distances, yet basketball demands a larger ratio of high-intensity running and sprinting, especially in populations other than elite males.

Basketball, volleyball, and handball require frequent jumping and basketball requires considerably more lateral movement than other sports.

1 Introduction

Within the category of multi-directional sports, the volume of demands between sports is widely variable. Certain multi-directional sports require frequent and consistent sagittal plane sprinting and high-intensity running [1, 2], while others demand more lateral shuffling or cutting, and some are heavily reliant on jumping [3–5]. Previous research has quantified sport-specific demands in various ways. The simplest method is observational time-motion analysis, which involves an individual or research team watching and counting the number of activities during live action or recorded video footage. While reliable, this method is time consuming, potentially variable, and unrealistic for real-time assessment of training volume or for comparison with other norms. Newer, wearable technology such as accelerometers, inertial measurement units (IMUs), or global positioning system (GPS) tracking sensors are beginning to provide more valid, reliable, and time-efficient measures that will help coaches, strength and conditioning professionals, and clinicians quantify and track the volume of sport-specific demands during play [6].

Quantification of sport-specific demands is important when developing team-based training and analyzing individual athletic performance. Pre-season training should be implemented with the intention of building up the players' level of conditioning to meet the demands of the sport at

the onset of competition, and allow athletes to avoid fatigue and overtraining, which may exacerbate injury risk and delay normal healing and recovery [7]. Recent investigations have been dedicated to achieving and maintaining optimal loading during short- and long-term training [7–10]. For example, the acute:chronic workload ratio inspects the rate of loading increase of sport-specific demands (generally running) to prevent performance declines and subsequent injury [7]. Ideally, these workloads should be dictated by the competition demands expected of athletes for their respective sports. Normative data available for sports for both sexes at varying ages can help coaches design appropriate and safe training programs, as well as equip trainers and rehabilitation specialists with the appropriate metrics for proper and safe return to competition following injury.

Normative sport-specific demands are crucial during the development of rehabilitation programs and should be part of the decision-making process when making return-to-play decisions [11]. Sports rehabilitation programs attempt to balance an athlete's expeditious return to play while reducing the risk of future injury and/or re-injury. Standard rehabilitation programs commonly progress from initiating strength and activation of local and regional musculature, to integrating neuromuscular control through controlled movements progressing from simple to complex, and concluding by optimizing biomechanics during sport-specific activities for safe return to play [12]. Rehabilitation programs designed to be sport-specific may be more effective if focused on meeting the activity demands of the athlete's sport; however, many current sport-specific programs simply incorporate the use of a ball, stick, club or other piece of athletic equipment to simulate the sport or emphasize a predominant physical demand of a sport (e.g., jumping in basketball). This approach becomes problematic if programs do not sufficiently address the specificity of the training principle that stipulates training is most effective when the activities closely resemble the sport demands in which improvements are sought [13]. It is the responsibility of clinicians to understand a rehabilitating athlete's sport-specific demands in order to create the most comprehensive rehabilitation and return-to-play program. Often times these factors are neglected or vastly underdosed, as they are not designed to accommodate the volume, frequency, or duration of sport demands such as the frequency of jumping and cutting, specificity of eccentric load, or the duration of sprinting that may be necessary for an athlete to safely return to play. This lack of congruency between rehabilitation and return-to-play programs and sport-specific demands may be partially responsible for high lower extremity (e.g., anterior cruciate ligament ruptures, hamstring tears, groin injuries) re-injury rates that occur in multi-directional sports [14].

While sport-specific demands have been independently quantified in small populations during a number of multi-directional sports, there are no comprehensive analyses synthesizing and comparing the demands across sports, age groups, or sexes. To progress toward more specific training development, performance analytics, rehabilitation programs, and return-to-play protocols, it is reasonable to expect that comprehensive, sport-specific analyses quantifying the lower extremity demands of multi-directional sports are warranted. Thus, the objectives of this systematic review were to characterize, quantify, and compare the demands of multi-directional sports. We hypothesized that there would be substantial differences in these variables across sports, ages, and sexes.

2 Methods

2.1 Study Design

A systematic review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) statement [15]. The protocol was not registered prior to initiation of the project and did not require Institutional Review Board approval.

2.2 Search Strategy

An electronic database search was conducted of the PubMed, CINAHL, SPORTDiscus, and Cochrane Central Register of Controlled Trials for all published manuscripts prior to April 2017. The following strategy was used for the search: ('time-motion analysis' OR 'physiological response' OR 'movement profiles' OR 'physical demands' OR 'competition demands' OR 'movement patterns') AND ('multidirectional sports' OR 'basketball' OR 'soccer' OR 'football' OR 'lacrosse' OR 'volleyball' OR 'field hockey' OR 'handball' OR 'futsal'). These sports were chosen based on their multi-directional demands and high rates of non-contact lower extremity injuries [16–20]. No filters or limitations were imposed during the search.

2.3 Eligibility Criteria

Studies were included in the review if they met the following criteria: (1) the study was written in English, (2) the study included any of the following multi-directional sports: soccer, basketball, lacrosse, handball, field hockey, futsal, or volleyball (beach or court) at any competitive level, (3) the study took place on a full-sized court/pitch with the regulation number of players, (4) the study included time-motion analysis data for the entire duration

of competition (practice data excluded), and (5) the study reported at least one of the following time-motion analysis variables: total distance covered, or characteristics of sprinting, high-intensity or high-speed running, jumping, lateral movement, changes in activity, and/or changes in direction.

2.4 Study Selection

The titles and abstracts of all studies obtained during the aforementioned search were reviewed independently by two researchers. All studies that at least one of the researchers deemed appropriate were retrieved for full-text review. The researchers each independently reviewed the studies and documented whether it was worthy of inclusion. Inter-rater agreement was calculated using a Fleiss κ test with values interpreted as <0: poor agreement; 0.01–0.20: slight agreement; 0.21–0.40: fair agreement; 0.41–0.60: moderate agreement; 0.61–0.80: substantial agreement; and 0.81–1.0: almost perfect agreement [21]. A third author was consulted for any disagreements about a study's inclusion.

2.5 Quality Assessment

A modified version of the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) checklist [22] was used to assess the quality and potential for bias of each individual study. Modifications to the STROBE checklist have been previously reported so that the quality assessment tool best assesses a study's research question [23]. The checklist used in this study comprised 11 items that are identified in the Electronic Supplementary Material, Appendix S1. The items addressed various aspects of the studies, from ample description of the participants and methodology, to sources of bias, and the transparent reporting of data. Two reviewers independently scored the included studies and formed a consensus for each score.

2.6 Data Extraction

Data describing the population studied were extracted from each study and organized based on sport and sex of the participants for ease of comparisons within and between sports. Additionally, based on competition level, data were organized into either elite (professional or international level), sub-elite, or junior (participants under the age of 18) levels.

Means and standard deviations (when reported) for the following straight-line activity demands were extracted from each study: total distance and characteristics of

sprinting and high-speed or high-intensity running, which included distance, mean distance, distance relative to total distance, duration, mean duration, and frequency. Total distance was defined as the summated distance an athlete covered over the course of competition, while total sprint and total high-intensity running distance were defined as the summated distance during which the athlete was at the defined speed for sprinting or high-intensity running in each study, respectively. Additionally, means and standard deviations for the following multi-directional activity demands were extracted from each study: frequency of activity changes, accelerations/decelerations (based on author-reported definitions), and characteristics of jumping, cutting, and lateral movement, including the frequency, distance, relative distance, and relative time. Meta-analysis was considered but ultimately deemed unrealistic due to the considerable variability of the definition of each variable.

3 Results

3.1 Search Results

Initially, the electronic database search yielded 966 citations. After duplicates were removed ($n = 272$), 694 titles and abstracts were reviewed. Because they did not meet the inclusion criteria, 562 were excluded, leaving 132 studies screened during full-text review. The research team's screening process exhibited almost perfect agreement (κ agreement = 0.89, 95% confidence interval [CI] 0.80–0.98), including 70 studies after full-text review. An additional hand search identified 22 studies that met the inclusion criteria, resulting in 92 manuscripts included in this systematic review (54 for soccer [1, 2, 24–75], 13 for basketball [3, 4, 76–86], 10 for handball [5, 87–95], 7 for field hockey [96–102], 4 for futsal [103–106], 4 for volleyball [107–110], and 0 for lacrosse). A complete view of the search and selection process can be found in the flow diagram in Fig. 1.

3.2 Quality Assessment

Results of the quality assessment using a modified version of the STROBE checklist showed a wide range of scores (5–10), with an average score of 8.2 out of 11. In general, the majority of studies showed adequate reporting of methodology and gave specific definitions of movement variables. However, only 63% of studies provided relevant dates of data collection, and nearly 94% of studies did not include an adequate description of inclusion/exclusion criteria, or failed to discuss the health and injury status of the athlete participants.

3.3 Sport Demands

Means and standard deviations for straight-line running demands, including total distance, characteristics of sprinting and high speed/intensity running, and the differences in these variables between sports, sex, and age can be found graphically in Figs. 2, 3, 4, 5 and 6, with raw data and study definitions presented in table form in the Electronic Supplementary Material, Appendices S2–S6. Means and standard deviations for multi-directional demands, including total activity changes, cutting, shuffling, and jumping frequency, and shuffling distance, and the differences between sports, sex, and age can be viewed graphically in Fig. 7. Additionally, raw extracted data, including other characteristics of these variables, can be found in table form in the Electronic Supplementary Material, Appendices S7–S12. A limited number of studies evaluated a diverse range of competitive levels (i.e., elite and sub-elite), which has been detailed in the Electronic Supplementary Material. Table 1 provides a summary table of median values for the most common and consistently defined sport-specific demands, including total distance, lateral movement distance, and jumping frequency.

3.3.1 Soccer

Twenty-one studies reported total distance travelled in male elite soccer players with the averages between 9000 and 12,000 meters (m) [1, 2, 24, 28, 30, 31, 41, 43, 49, 51, 53, 55, 62, 64, 66, 67, 73–75]. Total distance was generally lower and more variable in male junior soccer players, ranging from 2186 to 9900 m [27, 35, 37, 38, 40, 47, 53, 65]. In the five studies that reported on elite female soccer players, similar total distances to male elite soccer players were found, with a range from 9600 to 10,440 m [42, 48, 52, 56, 59, 71]. Similar to males, female juniors travelled less total distance during competition, ranging from 6900 m in the under-15 age group to over 8500 m in the under-17 age group [72].

Total distance travelled during sprinting was also highly variable, yet consistently larger in males than females and adult levels compared with juniors. Studies that used a velocity-based definition of sprinting reported ranges in elite males of 117–831 m [1, 24, 25, 28, 30–32, 36, 41, 43, 46, 49, 51, 55, 63, 66, 69, 74, 75] and studies that used subjective definitions of sprinting (e.g., maximal effort, movements with greatest extension of lower leg and highest heel lift) were more inconsistent, reporting distances of 194–1100 m [2, 40, 47, 75]. Total distances during sprinting were varied and ranged among groups; for junior males 114–325 m [27, 35, 37, 40, 47, 65], elite females 160–615 m [48, 52, 56, 59], and junior females 76–235 m [72]. The frequency of sprinting activities was

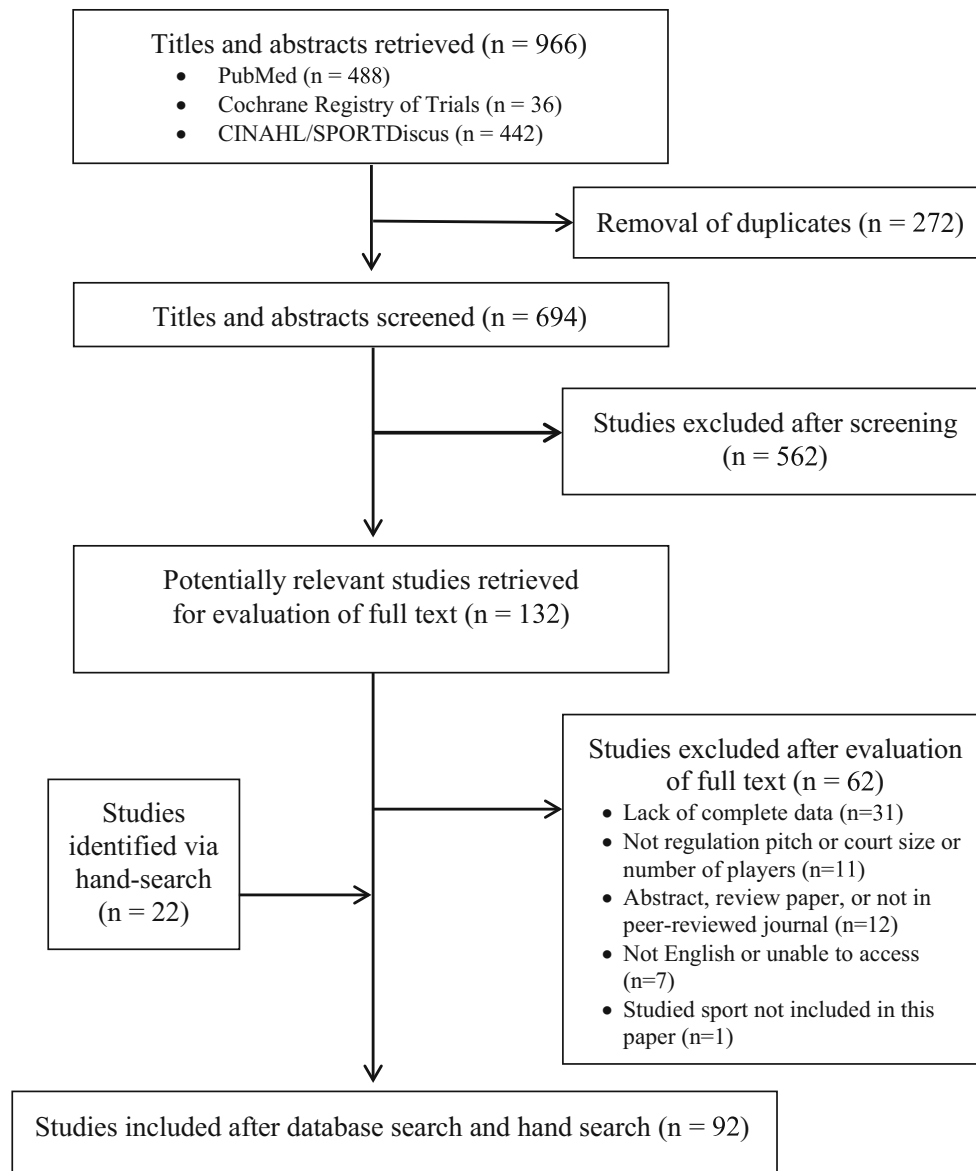


Fig. 1 Flow diagram of retrieved, screened, and included studies

less variable across ages and sexes, with elite males, junior males, and elite females all averaging 7–61 sprints [1, 46, 51, 52, 59–61, 63–65, 70, 73]. Junior female soccer players performed less frequent sprinting, averaging 5–13 sprints per game [72]. Other spatial and temporal characteristics of sprinting were also identified in some studies. For example, distance travelled while sprinting accounted for 4.0–12.3% of the total distance traveled [2, 67] and 0.3–3.4% of total game time [36, 44, 59, 65, 67, 73], totaling 34–192 s [1, 2, 36]. Each sprint averaged about 2.0 s in duration for 15.2 m in males and 19–29 m in females [34, 60, 70–72].

Soccer players generally spent more time and distance in high-speed/intensity running than sprinting (specific definitions used for sprinting and high-intensity running can be

found in the Electronic Supplementary Material, Appendix S2). On average, adult male soccer players traveled 222–1900 m in high-intensity running [1, 2, 24, 25, 28, 30, 32, 34, 41, 46, 49, 50, 53, 55, 63, 66, 69, 74], compared with 468–1740 m in male junior players [27, 35, 53, 65], 608–2452 m in elite female players [42, 48, 52, 56, 59], and 458–658 m in junior female players [72]. Nine studies reported the frequency of high-intensity running, with males averaging 69–168 [1, 58, 60, 63–65] and females averaging 125–376 high-intensity running activities per match [52, 56, 59]. In males, the distance travelled during high-intensity running accounted for 5.1–18.2% of the total distance traveled [2, 33] and 2.1–6.1% of total game time [2, 33, 58, 60, 62, 65], totaling 116–347 s [1, 2, 58]. Each

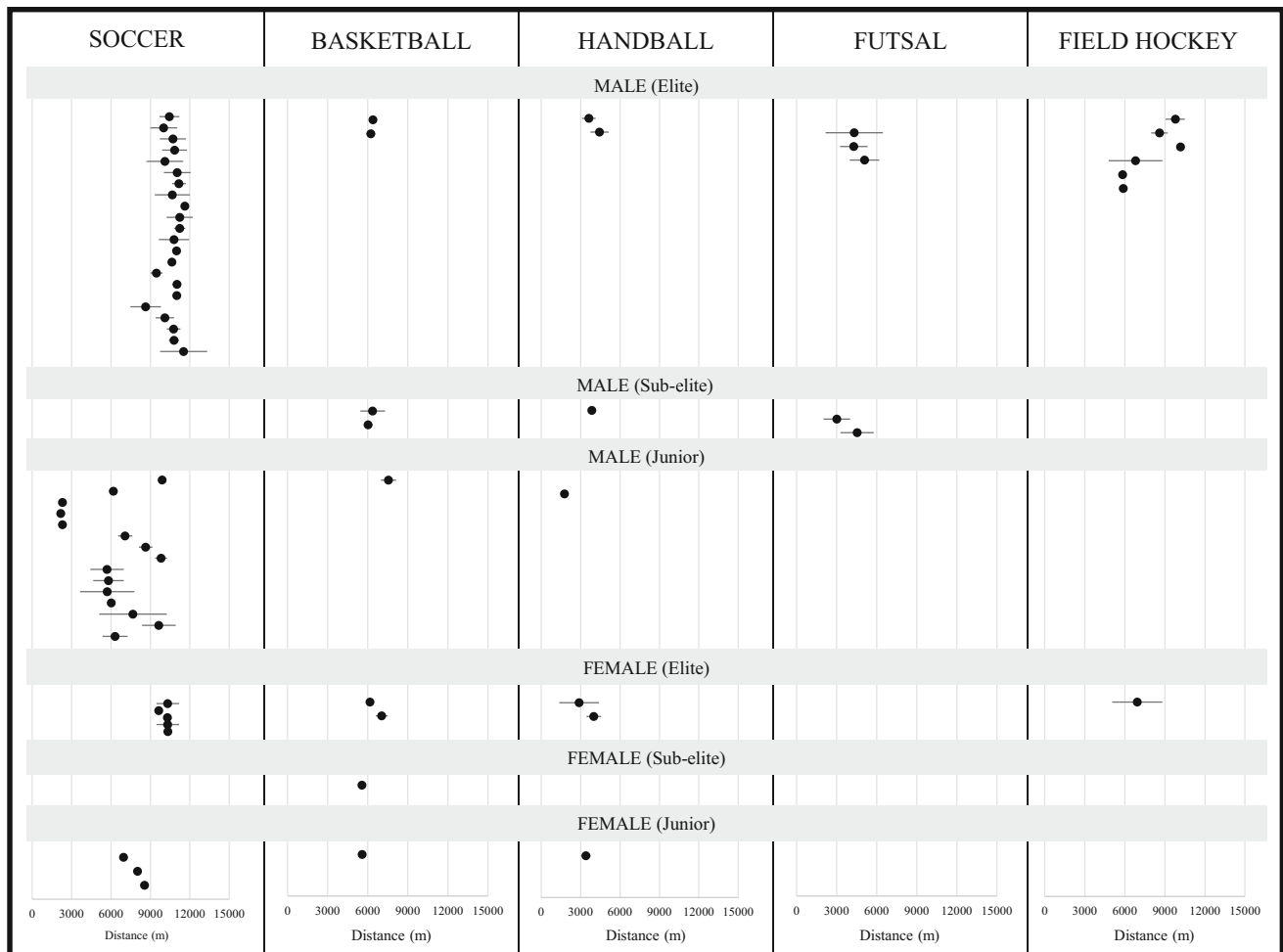


Fig. 2 Total distance traveled (mean $\pm$ standard deviation) over the course of a match in five multi-directional sports for male and female players at the elite, sub-elite, and junior level

high-intensity run averaged 1.3–4.4 s in duration [58, 65]. In females, high-intensity running accounted for 4.8% of the total time, with each high-intensity run averaging 2.3 s in duration over an average distance of 15.1 m [52, 71].

Over the course of a soccer game, both male and female players consistently averaged between 1379 and 1459 total activity changes [52, 59, 67], leading to one change every 4–4.5 s or 0.6 ± 0.2 changes per second [26, 31, 52, 59, 67]. The total number of accelerations $>2.5 \text{ m s}^{-2}$ ranged from 52 to 100 males [41, 55], and was considerably higher (423 per game) in elite females [57]. One study reported that high-intensity decelerations (16–32 per game) were considerably more frequent than high-intensity accelerations (4.8–8.0 per game), with midfielders performing the most and attackers the least [73].

Three studies reported the frequency of cutting over the course of a game [29, 61, 68]. Bloomfield et al. [29] performed a thorough cutting analysis, showing that elite male soccer players cut at angles $<90^\circ$ over 300 times to both the left and right over the course of the game. These

investigators also reported 45–49 cuts to each side at angles between 90° and 180° , two to three cuts to each side at angles between 180° and 270° , one cut to each side between 270° and 360° , and 7–8 ‘swerves’ in both directions [29]. Similar analyses, albeit using different categorizations, were performed by Robinson et al. [68], who reported 35–38 cuts in each direction between angles of 45° – 135° , and 20.2 ± 9.0 cuts at angles of $>135^\circ$. In general, midfielders were found to cut more often than both forwards and defenders [68]. Nedelec et al. [61], collecting only “hard changes of direction”, found that soccer players performed only 11.9 ± 5.2 such changes over the course of a game. However, these data are less generalizable and have limited application here due to the specificity of the outcome variable.

Soccer players were reported to travel laterally (shuffle) between 217 and 549 m throughout a game [35, 40, 75]. These demands accounted for 3.0–5.6% of the total distance traveled [40, 67], and 3.9–4.5% of the total time during a match [29].

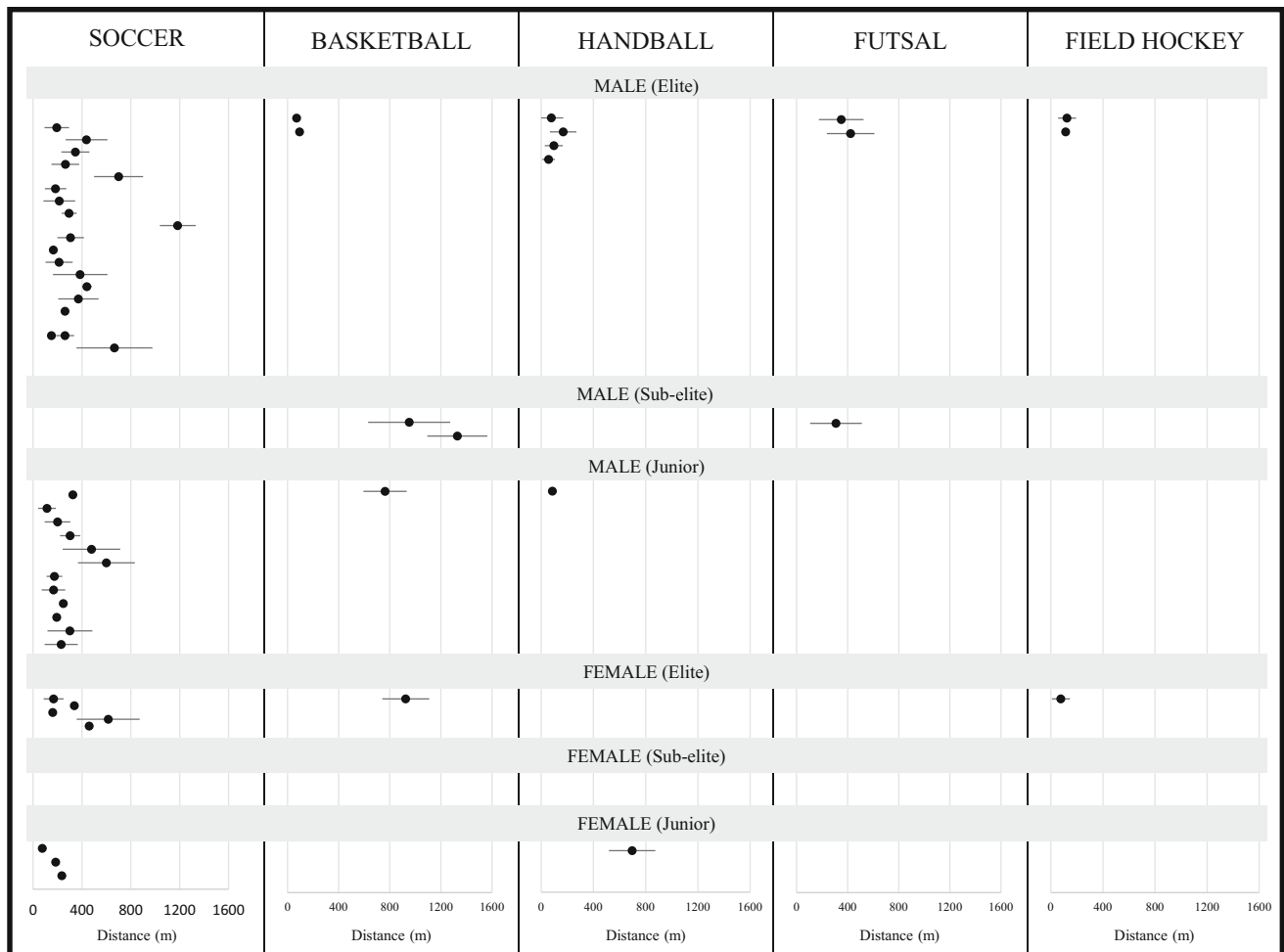


Fig. 3 Total distance traveled while sprinting (mean $\pm$ standard deviation) during competition in five multi-directional sports for male and female players at the elite, sub-elite, and junior level

Only two studies were found to report on characteristics of jumping in soccer players [61, 70]. On average, elite male soccer players were reported to jump 10.4 ± 5.4 times per game [61], while junior male soccer players performed 0.9–3.6 jumps per game [70].

3.3.2 Basketball

Significantly fewer studies have reported total distance covered during competition in basketball than soccer. However, across sexes, ages, and competition levels, straight-line running data are less variable, with basketball players averaging 6034–6390 m in adult males [4], 7558 m in junior males [77], 5576–7039 m in elite females [82, 84], and 5587 m in junior females [82].

Total distance travelled during sprinting was more variable, with substantially less distance reported in males than females. Elite males travelled 70–94 m over the course of the game [4], whereas sub-elite males, junior males, and elite females travelled 952–1329 m [4], 763 m

[77], and 925 m sprinting [84], respectively. Basketball players averaged 18–105 sprints [3, 4, 76, 78–81, 84], depending on the definition of sprinting ($>7 \text{ m s}^{-1}$ vs subjective definitions). Across all groups, distance travelled while sprinting accounted for 2–6% of total game time [76–79, 86], totaling 9–136 s [4]. Each sprint averaged 0.5–2.4 s in duration for 3.9–9.5 m [4, 78, 81, 86].

Minimal information has been reported on high-intensity running in basketball players. In one study of junior males, basketball players travelled an average of $406 \pm 109 \text{ m}$, which accounted for $2.4 \pm 0.5\%$ of the total playing time [77].

The number of activity changes that occur during a basketball game was reported in seven studies [3, 4, 76, 78, 79, 81, 84]. Considerable variability was identified in the data with ranges of 997–2733 per game in adult males [4, 81], 1050–1105 in junior males [77, 78], and 579–1750 in adult females [3, 79, 84]. McInnes et al. [81] reported a change in activity every 2.0 s, while Matthew and Delestrat [3] reported 21.2 activity changes per

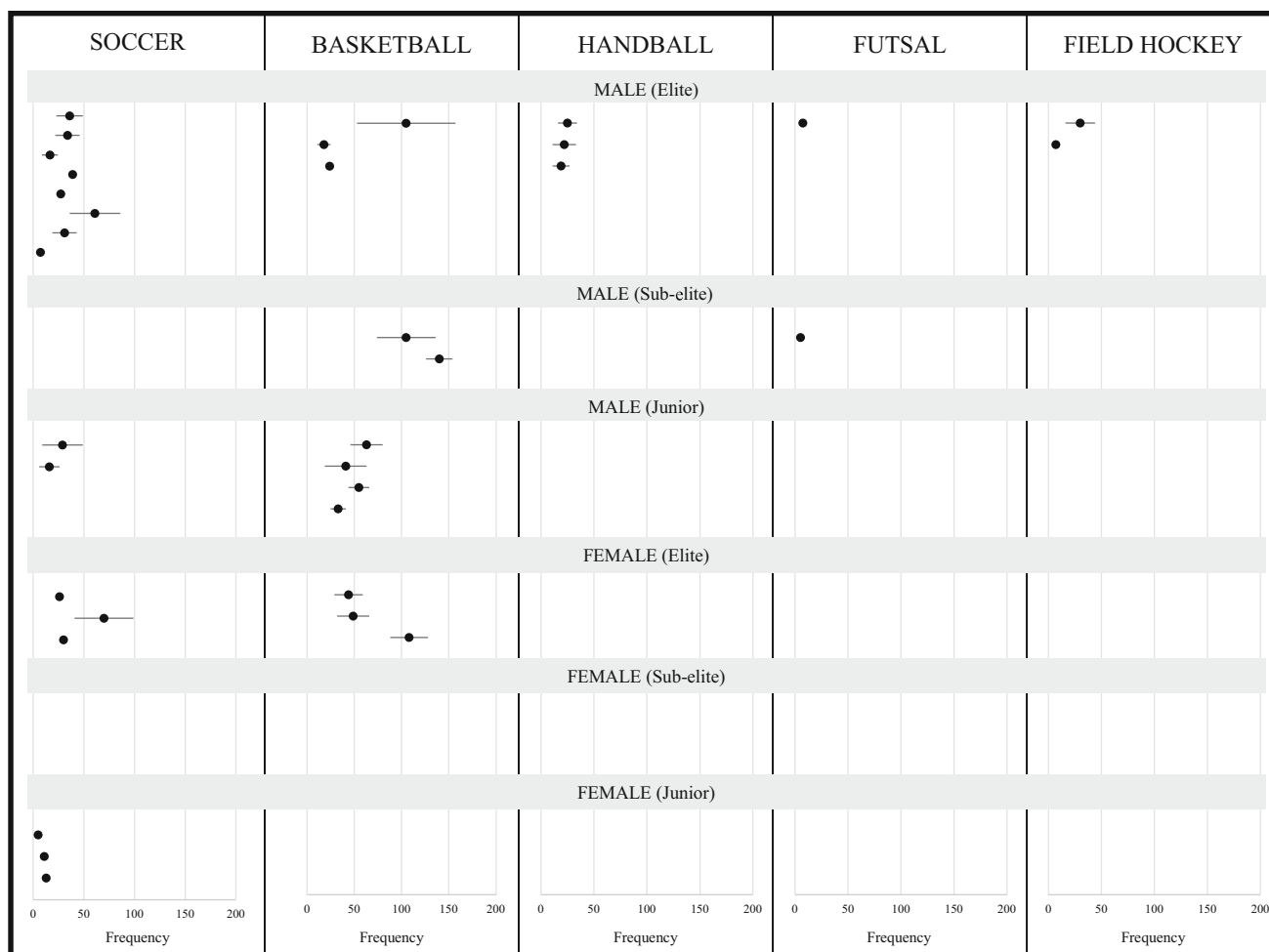


Fig. 4 Frequency of sprinting (mean $\pm$ standard deviation) during a match in five multi-directional sports for male and female players at the elite, sub-elite, and junior level

minute. Activity changes appeared to be relatively consistent across playing positions [4, 78, 84]. No studies were identified that reported the characteristics and/or frequency of accelerations, decelerations, or cutting over the course of a basketball game.

Characteristics of lateral movement were reported in ten of the basketball studies that were included in the review [3, 4, 76–81, 83–85]. The frequency of lateral shuffling was higher in basketball than the other sports, with more frequent shuffling in junior males (313–466 shuffles per game) than by adult males (91–345 shuffles) and adult females (63–298) [3, 4, 77–81, 84, 85]. Generally, more shuffles were reported to occur at moderate or low intensities than high intensities [3, 4, 76, 78, 79, 81, 83, 84]. Four studies reported the total lateral distance travelled during a game, with distances ranging from 203 to 269 m in adult males [4], 944 m in junior males [77], and 125 m in adult females [84]. The mean distance of lateral movements was reported in one study, revealing that distances were lower in elite (high-intensity: 1.88–2.23 m; low-

intensity: 1.24–1.31 m) than sub-elite males (high-intensity: 2.28–4.24 m; low-intensity: 1.52–1.60 m). Time spent in lateral movement was relatively high, accounting for 18.1–42.1% of game time [76–79], averaging between 0.62 and 1.42 s per high-intensity lateral movement and 0.84–1.98 s per low-intensity lateral movement [4].

The frequency of jumping appeared to be higher in males than females, with males averaging 41–56 jumps per game [4, 76, 78, 81, 85] and females averaging 19–43 [3, 79, 84]. Jumping accounted for 0.6–2.0% [76, 78, 79] of real time, averaging about 1.0 jumps per minute [79].

3.3.3 Handball

Total distance traveled during handball competition was reported in seven studies [5, 87–89, 91, 93, 95]. Male juniors were reported to travel 1777 m, the least of any age or competition level [89]. Across other levels, total distances were more consistent, with elite males travelling 3627–4440 m [5, 88], sub-elite males 3855 m [95], elite

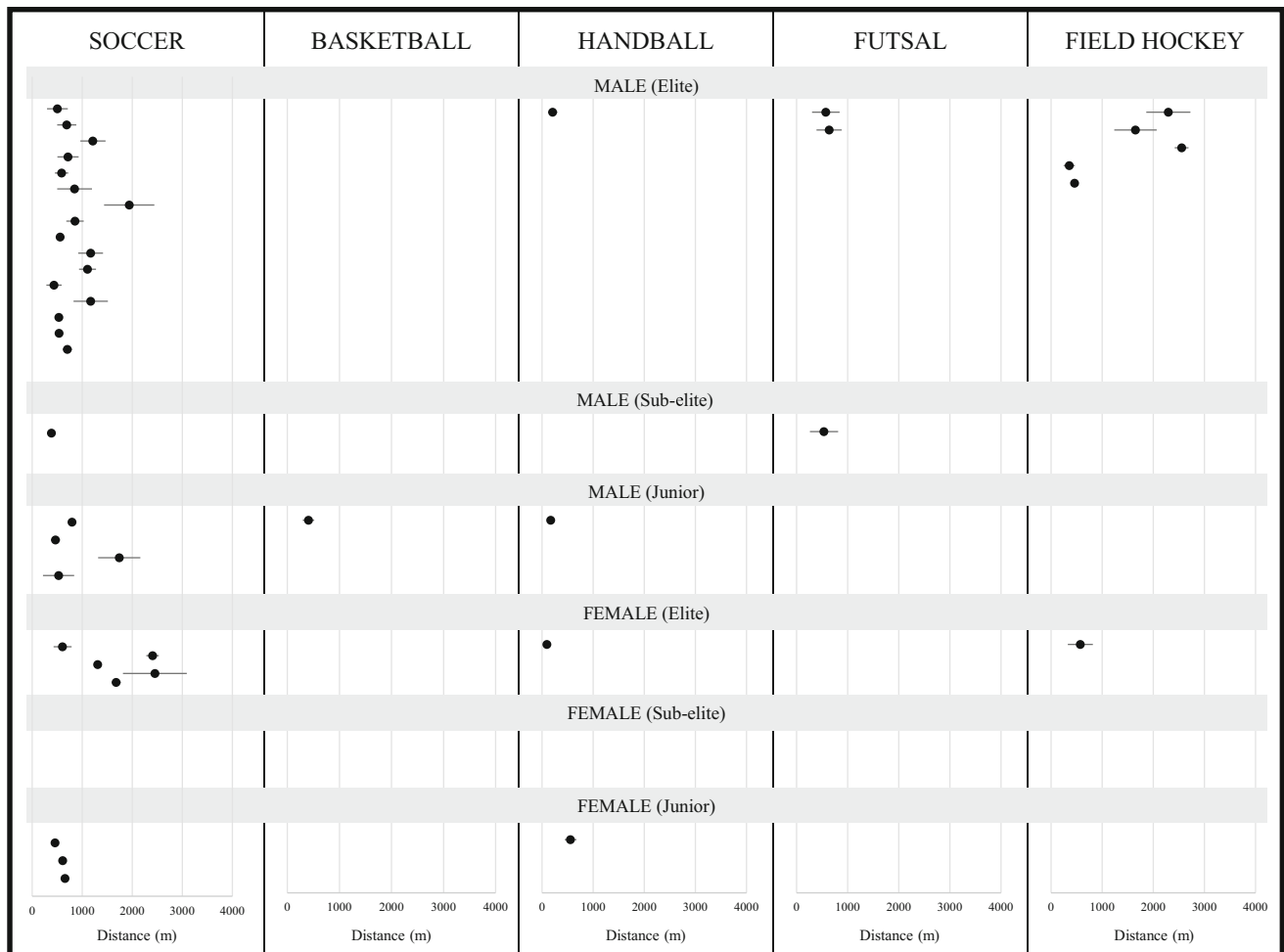


Fig. 5 Total distance traveled during ‘high intensity/speed running’ (mean $\pm$ standard deviation) during competition in five multi-directional sports for male and female players at the elite, sub-elite, and junior level

females 2882–4002 m [88, 91, 93], and junior females 3399 m [87].

Total sprint distance in male elite handball players ranged between 57 and 168 m [5, 92] and 86 m in male junior players during competition [89]. One study reported significantly larger sprint distances in junior female handball players, with 696 m accounted for by sprinting, although this study did not use a velocity-derived definition of sprinting [87]. Only one study reported sprint frequency, ranging from 19–25 sprints across three different handball positions in elite males [5]. Across sexes and ages, the distance travelled while sprinting during a game accounted for 1.5–3.9% of the total distance traveled [5, 87, 92, 95] and 0.1–0.6% of total game time [5, 88], totaling 2–26 s [5, 93]. Each sprint averaged 0.9–3.0 s in duration for 10.0–19.1 m [5, 91, 93].

On average, handball players changed activities 501–663 times per game, equaling an activity change every 4.6–5.9 s [89, 93]. Changes appeared to be dependent on position, as pivots changed activity (722.6 ± 110.9) more

often than wings (662.4 ± 97.4) and backcourt players (629.6 ± 79.7) [93]. Accelerations/decelerations performed in an entire game were not reported in any of the included articles, although Manchado et al. [91] reported 191 ± 71 total acceleration changes per minute in elite females throughout the course of a game using computer-assisted video analysis and no detection threshold. On the contrary, Luteberget and Spencer [90] used wearable accelerometers at a threshold of 2.5 m s^{-2} and reported 0.7 ± 0.4 accelerations, 2.3 ± 0.9 decelerations, and 1.0 ± 0.4 changes of direction per minute in elite females. In relation to position, one study [5] reported that on average, backcourt players (37.9 ± 9.2) changed direction more often than pivots (35.4 ± 11.1) and wings (18.4 ± 6.7).

Characteristics of lateral movement were reported in five handball studies [5, 88, 89, 92, 93], revealing that male handball players travel further in the lateral direction ($270\text{--}666 \text{ m}$) than female players ($138 \pm 99 \text{ m}$). On average, 15–55 lateral movements were reported over the

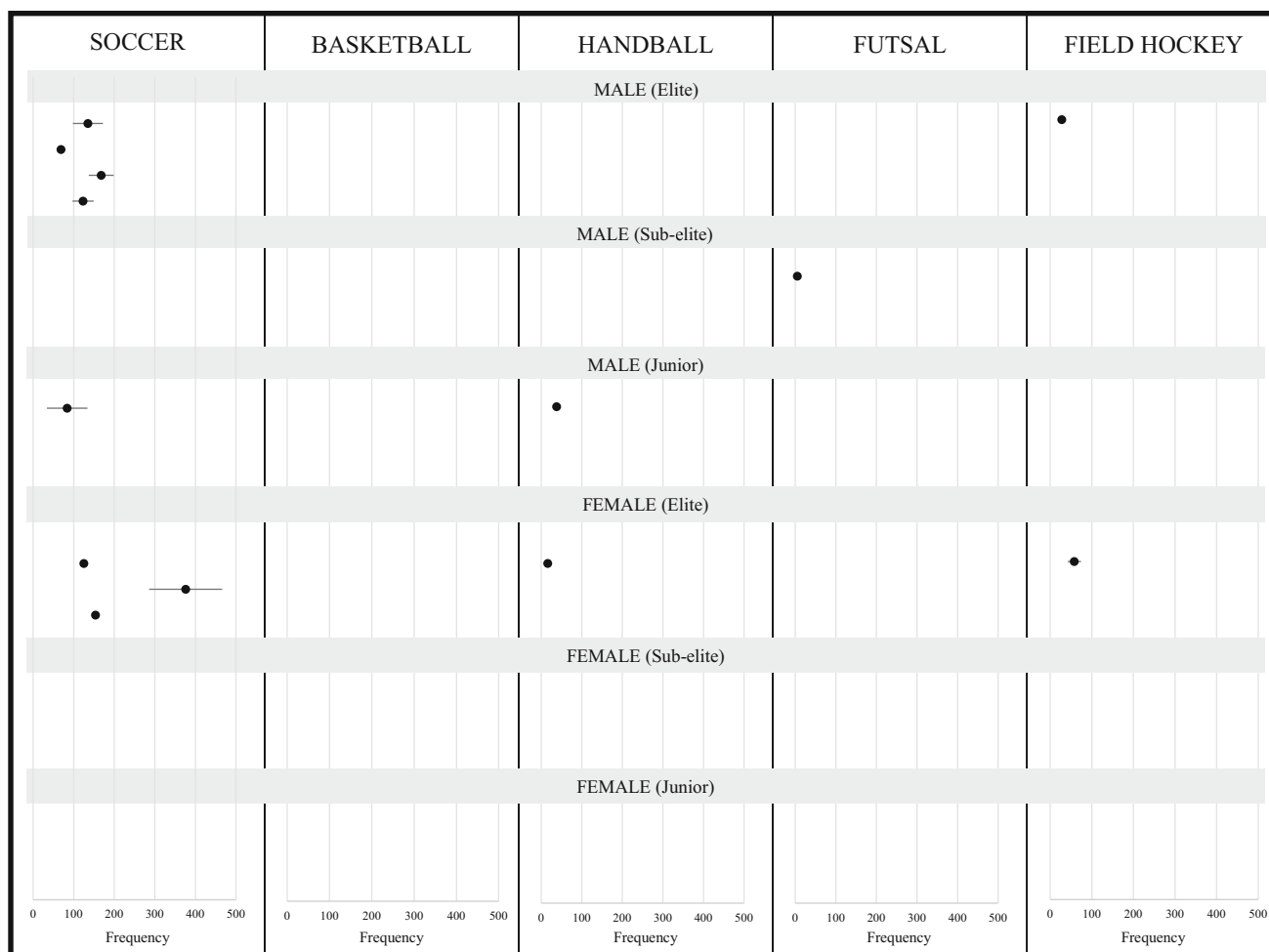


Fig. 6 Frequency of ‘high intensity/speed running’ (mean $\pm$ standard deviation) during a match in five multi-directional sports for male and female players at the elite, sub-elite, and junior level

course of a game, with wings (15 ± 13) and backcourt players (16 ± 12) moving less frequently than pivots (30 ± 19) [5]. This lateral movement has been reported to account for 0.7–7.4% [5, 88, 93] of real time and 18.4% of the total distance [92] travelled during a game. On average, each high-intensity lateral movement takes 2.0–2.7 s and covers 7.7–10.4 m.

Two studies have reported on the characteristics of jumping in handball players, with very inconsistent results [5, 89]. Povoas et al. [5] reported that elite male handball players jump 8.2–19.1 times per game, with backcourt players jumping more frequently than pivots and wings. In a study of junior male handball players, Chelly et al. [89] reported 42.8–45.9 jumps per half, with more in the first than second half of a competition.

3.3.4 Futsal

Of the four studies on futsal players, the only data presented were on adult male players [103–106]. Total

distance covered during competition was consistent between groups, with elite males playing outfield positions travelling 4277–5087 m, and sub-elite male outfield players travelling 3011–4528 m per game [103–106].

Distance travelled while sprinting was reported to be 308–422 m in male athletes [103, 105, 106], accounting for approximately $8.9 \pm 3.4\%$ of the total distance [103], and 5.6–6.5% of the total time [106]. Elite male futsal players performed 7.4 ± 3.3 sprints per game, which was slightly more than the 5.3 ± 4.1 sprints in sub-elite males [104]. On average, sprints lasted for 1.6–1.9 s over a distance of 7.8–13.0 m [104]. Elite male futsal players travelled an average of 534–636 m in high-intensity running [103, 105, 106], which accounted for 13.7% of the total distance traveled and 7.7–8.7% of the total game time [103, 106].

Only one study was identified that reported the multi-directional activity demands in futsal [104]. On average, futsal players total 306–468 activity changes per game, with elite players changing activity more frequently than

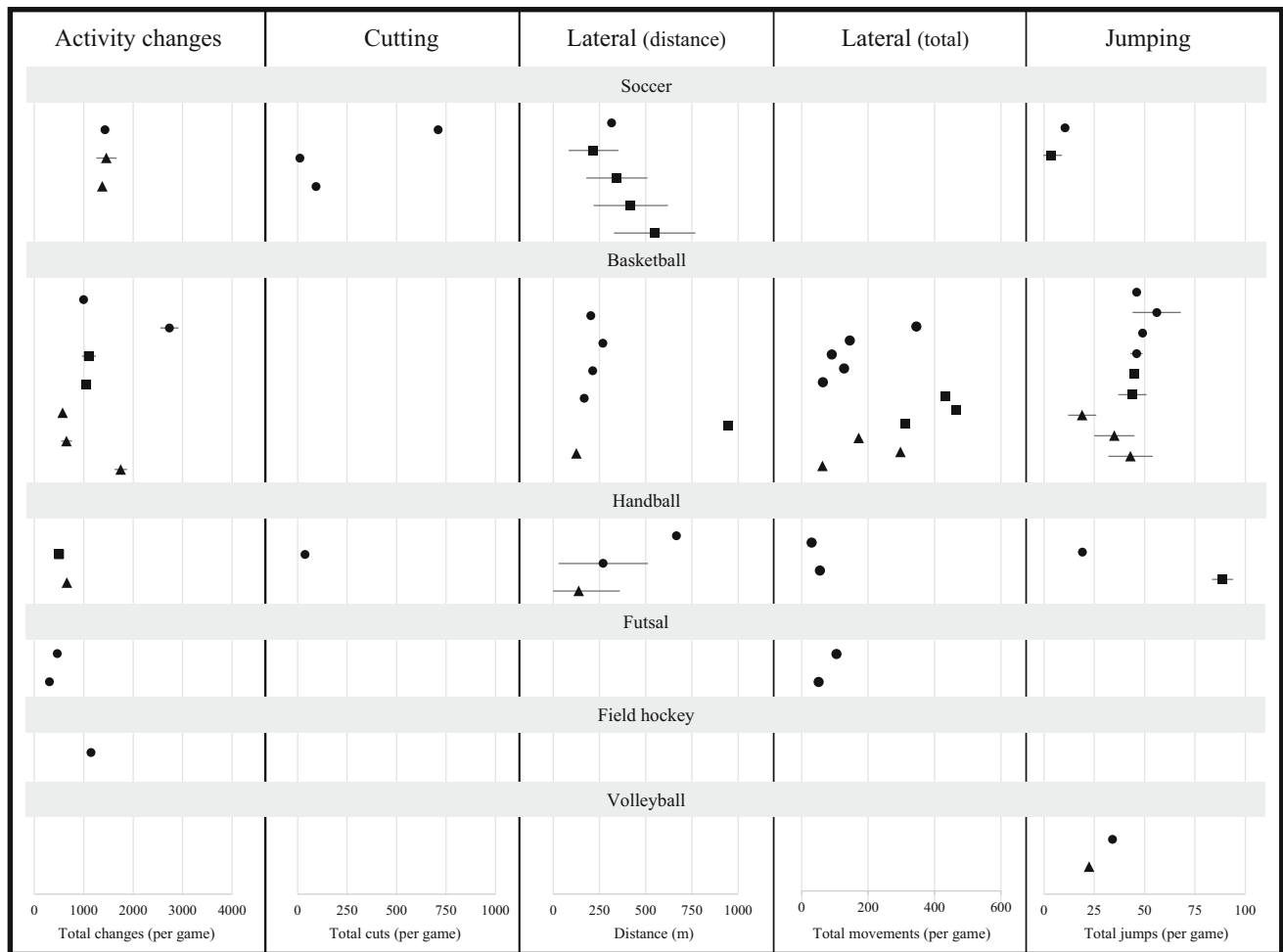


Fig. 7 Total distance traveled (mean $\pm$ standard deviation) over the course of a match in five multi-directional sports for male adult (*circle*), male junior (*square*), and female adult players (*triangle*)

Table 1 Summary table representing the median value of extracted sport-specific demand data for each category

Movement category	Player type	Soccer	Basketball	Handball	Futsal	Field hockey	Volleyball
Total distance (m)	Male	10,794	6300	3855	4313	7694	
	Female	10,321	6177	3442		6931 ^a	
	Junior male	6175	7558 ^a	1777 ^a			
	Junior female	8024	5587 ^a	3399 ^a			
Lateral movement distance (m)	Male	316 ^a	208	468			
	Female		944 ^a	138 ^a			
	Junior male	381	125 ^a				
Jump frequency	Male	10 ^a	48	19 ^a			34 ^a
	Female		45				23 ^a
	Junior male	4 ^a	35	89 ^a			

^a Based on one included study in this category

sub-elite players [104]. The same study also reported on the characteristics of lateral movement in male elite and sub-elite futsal players, reporting that futsal players travel

between 51 and 105 m in the lateral direction, with elite players travelling further than sub-elite players [104]. Additionally, these movements were reported to average

9.6–11.0 m and take an average of 2.6–3.2 s [104]. No data were presented on acceleration/decelerations, cutting, or jumping in futsal.

3.3.5 Field Hockey

Of the seven field hockey studies, six provided data on elite male field hockey players [97–102] and one provided data on female hockey players [96]. Elite males were reported to cover a broad range of distances (5824–10,160 m) [97–99, 101, 102] and females were reported to cover a more narrow range (6154–6931 m) [96].

Similarly, males covered more distance sprinting than females, averaging 114–124 m [99, 102] compared with 77 m for females [96]. Elite males averaged 7–30 sprints per game, accounting for 0.2–1.5% of the total time, with each sprint lasting for an average duration of 1.8 s [100–102].

The total distance covered during high-intensity running was quite variable in elite males, with two studies [97, 98] reporting distances between 1652 and 2554 m and two others [99, 102] reporting between 355 and 457 m. Elite males were reported to perform 28 episodes of high-intensity running per game, which accounted for 1.2% of the total game time [101, 102]. Elite females were reported to cover 369–571 m across 43–58 episodes of high-intensity running, dependent on playing position [96].

Similar to futsal, only one study was identified that reported on the multi-directional activity demands of field hockey players. Lythe and Kilding [99] reported that elite male field hockey players perform 1148 ± 128.9 activity changes per 70 min of game action. No data were presented on acceleration/decelerations, cutting, lateral movements, or jumping in field hockey.

3.3.6 Volleyball

No studies were identified in volleyball players that described the straight-line running demands of the sport; however, four studies were identified that quantified the multi-directional demands, two of which detailed characteristics of lateral movement in beach volleyball players, and two of which detailed characteristics of jumping in court volleyball players [107–110]. Lateral movements accounted for 67.8% of offensive movements in elite male beach volleyball players and 50.9% in female players [107, 108]. On defense, males also performed more lateral movements, accounting for 11.3%, compared with 6.9% of total movements in female players [107, 108].

One study reported that women's court volleyball players performed 22.5 jumping activities per game [110]. While a total number of jumps was not reported for male court volleyball players, it was reported that male court

volleyball players performed 1–19 block jumps, 0–15 spike jumps, and 0–21 jump sets per game, depending on playing position [109]. Jumping and landing patterns of female court volleyball players showed that when on offense, players mostly jumped off two legs (84% of jumps) as opposed to one (left 14%, right <1%), but landed with a more mixed pattern (bilateral 55%, left leg 35%, right leg 10%) [110]. On defense, players overwhelmingly jumped off both feet (99% of jumps), but had more varied landing patterns (bilateral 57%, left leg 16%, right leg 27%) [110].

3.3.7 Between-Sport Comparisons

Comparisons of demands between sports are visually depicted in Figs. 2, 3, 4, 5, 6 and 7, and median values are reported for comparison in Table 1.

4 Discussion

This systematic review provides normative data on the sport demands for straight-line running (sprinting and high-intensity running) and multi-directional (cutting, jumping, lateral movement) movements required of sexes and age groups across a number of team sports. As expected, there were substantial differences among sports and across the ages and sexes that should be considered when developing sport-specific training, rehabilitation, and injury prevention training programs. For example, sizeable differences were found in total distance traveled, with soccer and field hockey demanding the largest volumes, yet basketball demanded a larger ratio of high-intensity running and sprinting, especially in populations other than elite males. While the dataset presented in this review is robust and can be immediately applied to training and clinical settings, it also highlights the distinct lack of literature in some sports (lacrosse, volleyball) and inconsistent reporting of sport demand variables. The definition of sprinting was highly variable, ranging in definition from the low $>19 \text{ km h}^{-1}$, to the high $>30 \text{ km h}^{-1}$, and with others who defined sprinting based on effort or biomechanics. Some studies defined sprinting differently for males and females. As wearable technology continues to become more prevalent in practical use and research [6], standardized reporting procedures will be needed to better and more consistently analyze and compare data. Nevertheless, data presented in this review provide normative values for strength and conditioning professionals and rehabilitation clinicians and can serve as a launching point for future research.

Although all of the sports discussed in this review are considered multi-directional sports, straight-line running demands were the most commonly studied game demand, especially in soccer, futsal, and field hockey. Soccer

demanding the largest volume of running, ranging from 9000 to 12,000 m per game, nearly two times as much as basketball, handball, and futsal. This is not surprising given the size of the playing fields and greater surface area to player ratios in these sports. Not surprisingly, adult athletes traveled further than junior athletes across sports, with minimal differences in total distance between males and females. Similar patterns were not identified for the sprinting demands, as sprinting frequency tended to be highest in basketball but distances (despite differences in playing surface area) were consistent across sports. However, elite male basketball players covered significantly less distance while sprinting than sub-elite male, junior, and female basketball players. This may be a function of tactical differences in game play, considering total distance covered was similar between groups and elite males spent significantly more time in moderate-intensity activities [4]. Field hockey, particularly in males, showed the longest distances of high-speed running.

The characteristics of straight-line running, whether it is total distance, mean distance per bout, or time between bouts, need to be accounted for in performance training and rehabilitation from injury. Though anecdotal, many running programs suggest a '10% rule' for running progressions, where the distance increases are not >10% in any given week; a lack of safe progressions can potentially lead to overuse injuries, such as stress fractures or tendinopathies. More recent evidence has been linked to the acute to chronic workload ratios that compare an athlete's weekly load (acute) to their load over the prior month (chronic). An acute to chronic workload of 0.8–1.3 has been suggested, with data supporting ratios of >1.6 considered too high, placing athletes at a 3–5 times greater risk of sustaining an injury [111, 112]. Considering that the total distance travelled during some sports exceeds 10 km per game, safe systematic progressions are needed. Achieving such distances may require weeks, particularly following injury-induced lay-offs, and does not account for the time needed for sport-specific demands or unexpected delays in training or rehabilitation due to muscle soreness, recovery and tissue healing, and other concerns.

Compared with the number of studies that have detailed sagittal plane demands, far fewer studies have characterized the multi-directional demands of the sports included in this review. Activity changes were generally more frequent in basketball (every 1–2 s) than soccer (3–4 s), field hockey (3–4 s), and handball (5–6 s). Only three of the 92 studies included addressed the characteristics of cutting, all of which were in soccer populations. We anticipate that more robust analyses of cutting, accelerating, and decelerating will be studied in the future as the use of tri-axial accelerometers becomes more common. Quantifying these activity changes is important because repetitive

acceleration and deceleration places considerable demands on the lower extremity. More specifically, activities requiring frequent decelerations, as seen with repetitive cutting, may contribute to acute, non-contact, and overuse lower extremity injuries, including sprains and strains, secondary to the injury-inducing large forces created during eccentric loading of joints and soft tissue [113–115]. Cutting maneuvers place large concentric demands on the entire kinetic chain, especially the gluteal and quadriceps muscle groups for power generation, with the initial deceleration phase placing even larger eccentric demands on these muscle groups [116, 117]. Specifically, the magnitudes of abduction/adduction and internal/external movements at the knee vary significantly across various sub-phases of stance (e.g., weight acceptance, peak push-off, and final push-off) while performing sagittal plane movements such as straight-line running. Thus, cutting places large tri-planar loads on the lower extremity, which, if not purposefully trained for during rehabilitation, may lead to future re-injury.

The highest frequency of lateral movement demands was found in basketball, with upwards of 300 lateral movements per game reported in some studies. Matthew and Delestrat [3] reported that basketball players perform more purposeful movements in the frontal plane (shuffling) than the sagittal plane (running and jumping). Quantifying these movements may be extremely important because of the unique biomechanical demands associated with movements in the frontal plane. Most studies included in this review did not report clear definitions of lateral motion, which may differ between sports. For example, lateral movement in field sports may involve quick lateral cuts with limbs crossing the midline, while in basketball lateral movement is typically associated with shuffling. In itself, lateral shuffling is a highly variable movement [118] that is dependent on both the intensity of the movement and the frequent 180° direction change. Additionally, the primary direction of jumping and landing movements can occur in the frontal plane, which elicit larger peak hip internal rotation and knee external rotation angles, higher peak knee abduction and internal rotation external moments, and gross movement patterns that are largely unrelated to how an athlete lands during a sagittal plane landing [119]. As new training, rehabilitation, and prevention programs are designed, incorporating an adequate volume and frequency of lateral movement must be considered because of the significant volumes and distinct biomechanical demands associated with these types of movements. Thus, rehabilitation programs that strongly emphasize sagittal plane (forward/backward) movements without adequately simulating frontal plane/lateral demands may not adequately prepare athletes for safe progression and return to competitive action.

Basketball, volleyball, and handball were found to require the largest jumping demands, which should not be surprising considering that jumping and landing are the main causes of lower extremity injuries in these sports [120–124]. Of the activity demands characterized in this review, jumping may be the one variable in which the load has been adequately prescribed in training, rehabilitation, and injury prevention. Plyometric activities are common in these programs, and the volume may be adequate to reach the approximately 50 jumps that are required over the course of a basketball or volleyball game. Although we have a clear understanding of jumping loads in these sports, we still do not have a clear understanding of the characteristics of these jumping and landing patterns, and therefore cannot perfectly simulate each sports' specific demands. Tillman et al. [110] provide excellent detail regarding the jumping and landing patterns of elite women's court volleyball players, noting the variety of double- and single-leg jumps and landings that occur over the course of a match. However, this type of analysis has not been performed in basketball or handball, which is important considering that the landing pattern can influence the risk of subsequent injury. Landing on two legs elicits less risky biomechanics than single-leg landings by promoting greater hip and knee flexion angles; lower peak hip adduction angles; and lower peak hip flexion, adduction, internal rotation, and knee flexion, abduction, and external rotation moments [119]. High knee flexion moments observed during single-leg landings may cause overload to the patellar tendon, leading to tendinopathy, a common injury in both basketball and volleyball [125]. Additionally, higher knee abduction moments have been reported to place athletes at a higher risk for serious knee injuries, such as anterior cruciate ligament ruptures [126]. Thus, while current programs may adequately simulate the correct volume of jumping and landing during these multi-directional sports, more research is needed to better understand the landing patterns associated with these sports.

4.1 Clinical Application

Common training, rehabilitation, and injury prevention programs may simulate the technical aspects of sports and the distinct biomechanical and physiological demands of jumping, cutting, and lateral movement, but fall short of the load required during true competition. Training and rehabilitation programs are generally repetitive in nature, performed in isolation of other activity demands and for relatively short bursts, and do not account for the frequency or volume of activity changes that require acceleration/deceleration components. Our compiled data indicate that multidirectional sport athletes change activity 500–3000 times over the course of a competition, or approximately

once every 2–4 s. The period of transition from one activity to the next requires some level of acceleration and deceleration resulting in a higher risk of injury because of the high eccentric demands placed on the lower extremity [123, 127]. Thus, training and rehabilitation programs should simulate both the frequency and volume of sport demands with interval programs that start in a controlled environment and progress towards integration of reactive and unanticipated environments with other players and defenders. Traditionally, rehabilitation programs transition from one exercise to the next after a desired number (i.e., 6–12) of repetitions have been performed, and physical performance and biomechanical tests measure movement during only one type of activity, such as sprinting, jumping, or landing. While this local treatment approach may be logical early in the rehabilitation process, failure to incorporate skill components of the sport, such as unanticipated cutting, external feedback, and maximum effort, falls short of a global treatment approach. A paradigm shift may be necessary to deemphasize repetitive and isolated training progressions and promote integration of an assortment of sport-specific activities in to late-stage rehabilitation programs and return-to-play testing procedures. For example, the Functional Lower Extremity Evaluation (FLEE) battery [128] is a 45-min screening procedure that encompasses a number of screening tools such as the Lower Extremity Functional Test (LEFT), which progresses an athlete through a number of different agility demands performed one after another, effectively increasing the physiological and psychological demands required as the test progresses [129].

Clinicians and researchers have advocated for comprehensive return-to-play decision-making processes that include information on an athlete's physical and psychosocial status [130]. One such model is the Strategic Assessment of Risk and Risk Tolerance (StARRT) framework, which identifies the restoration of sports-specific skills and function and the restoration of the musculoskeletal system, and cardiopulmonary and psychological functions as primary determinants for return-to-sport decision making [131]. The StARRT model also accounts for the type of sport, position, and competition level, but makes no explicit mention of sport-specific demands. This systematic review attempted to quantify and report the characteristics of straight-line running, cutting, jumping, and lateral movement that various multi-directional sports require across sexes and ages for additional consideration with this model. This information provides the rehabilitation specialist with information regarding sport-specific expectations and achievement of sport-specific landmarks prior to return to play.

Armed with this knowledge, sports medicine professionals can now safely and confidently understand the

sport-specific volume to accurately deliver the correct dosage during training, rehabilitation, and injury prevention programs. For example, a program preparing a male soccer player to compete in a match should approach at least 10,000 m of total distance, 30 sprint repetitions totaling up to 300 m, 100 high-intensity running repetitions totaling up to 1000 m, and 300 m of lateral movement. These benchmarks can help reduce the chance of over- or under-dosing volume and load as athletes navigate the course of these programs. As in medicine, the specific dosage of a medication is crucial to the effectiveness of the treatment. As such, the sport-, competitive level-, and sex-specific data presented in this paper will complement the acute to chronic workload model [9, 112], highlighting the importance of gradually and systematically increasing workloads with an end goal that is similar to game demands, and thereby minimizing the risk of injury while simultaneously maximizing performance.

4.2 Limitations

This systematic review gathered data from a large number of studies, creating a dataset to guide clinicians in the development of future programs. However, there were limited data on some sports and considerable variability in the methodology, including differences in technology and definitions of activity that may have biased certain numbers. For example, some of the reported data were measured using GPS technology; however, sampling frequencies ranged from 1 to 10 Hz and frequencies <10 Hz have been suggested to give questionable and inconsistent results for accelerations and direction changes [132]. Variability of the results within and between sports may also be attributed to different rules of play that could influence tactical and physical demands.

We still know very little about men's and women's volleyball, field hockey, and lacrosse, even though their popularity rivals that of soccer and basketball in some locations. Additionally, considerable research has been performed in rugby and Australian Football, yet these sports were not included in this study. Data included in this review may provide an incomplete view of an athlete's load, considering that only competition data was published. Minimal studies have quantified load during practice, which in some instances may require even higher demands than those presented in this review and more research is necessary to gain a better understanding of the demands of technical training and team practices, where more repetition tends to occur, ultimately resulting in chronic and overuse pathologies. While we believe the clinical utility of these data to be high, it is important to note that many other non-modifiable individual-specific factors, such as past injuries, pathology, healing, psychological condition, and

playing position, will impact rehabilitation and return to sport. Professionals need to use the reported values as reference numbers, but tailor their training and rehabilitation programs on a case-by-case basis.

Lastly, as with all systematic reviews, the quality of the data reported in the review can only be as good as the quality of the included studies. In this particular review, the quality of some included studies was relatively low. Data may have been influenced by the variability of methodology and poor reporting of the health status of the athletes analyzed in each study. Participating athletes dealing with injuries at the time of collection could modify intensity levels, and subsequently alter demands required during competition.

5 Conclusion

Multi-directional team sports each consist of a unique combination of demands that define an 'activity profile' for each sport. Considerable variability exists in the demands of straight-line running, lateral movement, cutting, and jumping across sports, ages, and sexes. These data can be used to reinforce the specificity of training principle (including activity type and dosage) in training, rehabilitation, and injury prevention programs. Sports medicine professionals now have more information about these activity profiles and the interplay between volume and capacity, allowing them to make appropriate clinical decisions based on scientific evidence.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the conduct of this review or preparation of this article.

Conflicts of interest Jeffrey B. Taylor, Alexis A. Wright, Steven L. Dischiavi, M. Allison Townsend and Adam R. Marmon declare that they have no conflicts of interest relevant to the content of this review.

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